Jinwoo Jung

M.S. student, Hanyang University | ASML Lab, advised by Prof. Joon-Hyuk Chang

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Research Interests

Diffusion and flow-based generative models: controllable generation and editing, and training-free inference efficiency — applying only the edit a user requests while leaving the rest unchanged, and allocating sampling compute only where it is needed.

Education

Hanyang University, Seoul, Republic of Korea — Mar 2025 – Feb 2027 (expected)
M.S. in Artificial Intelligence Semiconductor Engineering

- Advisor: Prof. Joon-Hyuk Chang (Acoustic, Speech Signal Processing and Machine Learning Lab).

Konkuk University, Seoul, Republic of Korea — Mar 2019 – Feb 2025
B.S. in Computer Science and Engineering

- GPA 3.88/4.50 (major 3.97/4.50).

Zurich University of Applied Sciences (ZHAW), Winterthur, Switzerland — Sep 2022 – Jan 2023
Exchange student, Computer Science

Publications

* denotes equal contribution; † denotes corresponding author.

SARA: Step-Adaptive Rank Adjustment for Diffusion Inference Acceleration — NeurIPS 2026
Dong Hoon Lim*, **Jinwoo Jung***, Won-Gook Choi, Joon-Hyuk Chang

Edit the Moment, Keep the Rest: Time-Localized Audio Editing via Instruction — INTERSPEECH 2026
Jinwoo Jung, Gihun Son, Won-Gook Choi, Joon-Hyuk Chang [demo]

Branch-wise Complementary Attention for Acoustic Scene Classification — INTERSPEECH 2026
Seung-Gyu Han*, **Jinwoo Jung***, Pil Moo Byun, Won-Gook Choi, Joon-Hyuk Chang

Machine Learning Algorithm-Based Prediction Models for Adolescent Suicide Attempt — IEEE ICCE 2025
Jinwoo Jung, Seulmi Kang, Myeonghwan Go, Sieon Lee, Seokheon Cho

Manuscripts Under Review

Venue names are omitted while review is in progress.

ENCORE: Training-Free Few-Step Diffusion Sampling with Terminal Moment Matching — under review, 2026
Jinwoo Jung, Dong Hoon Lim, Joon-Hyuk Chang†, Defang Chen

TRACE: Trajectory-Conditioned Editing of Sound Sources in First-Order Ambisonics — in preparation
Jinwoo Jung, Dong Hoon Lim, Joon-Hyuk Chang

Presentations

SARA: Step-Adaptive Rank Adjustment for Diffusion Inference Acceleration — Dec 2026 (upcoming)
NeurIPS 2026, Atlanta, USA (poster; presenting author)

Edit the Moment, Keep the Rest: Time-Localized Audio Editing via Instruction — Sep 2026
INTERSPEECH 2026, Sydney, Australia (presenting author)

Branch-wise Complementary Attention for Acoustic Scene Classification — Sep 2026
INTERSPEECH 2026, Sydney, Australia

Patents

Branch-wise Complementary Attention for Acoustic Scene Classification — filed Oct 2026
Republic of Korea; examination requested

System and Method for Accelerating Diffusion Model Inference Using Step-Adaptive Low-Rank Adjustment — in preparation
Republic of Korea

Interval Conditioning and Source-Event Masking for Instruction-based Time-Localized Audio Editing — in preparation
Republic of Korea

Research Experience

Hanyang University, Seoul — Mar 2025 – present
ASML Lab, Adv. Prof. Joon-Hyuk Chang

- Ongoing collaboration with Defang Chen (postdoc, UC Berkeley) on training-free few-step sampling.

Qualcomm Institute, UC San Diego, La Jolla, USA — Jan 2024 – Feb 2024
Artificial Intelligence (AI) Development Project, 12th cohort

- Team lead; our team placed first and received the Outstanding Achievement Award.
- The team project continued after the program and became a conference paper (IEEE ICCE 2025), co-authored with the project's principal investigator.

Industry research projects, Hanyang University — 2025 – 2026

- AI Model Compression and Acceleration, with Samsung Electronics (Jan 2026 – Sep 2026).
- Sound-Tracking AI Agent via Connected Devices, with LG Electronics (Jun 2025 – Sep 2025).

Awards

1st Place, IC-PBL+ Research Showcase, Hanyang University — Jun 2026
Outstanding Achievement Award, Qualcomm Institute AI Development Project, UC San Diego — Feb 2024

Skills and Languages

Programming Python, PyTorch, librosa, torchaudio | **Languages** Korean (native), English (TOEFL iBT 100)